

Endpoint Coordinates and Wilder's Log-Odds RSI

Exact Geometry, Effective Length, and Finite-Horizon Normalization

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September 2026

Abstract

This paper develops an exponentially weighted endpoint coordinate for a time series and identifies Wilder's Relative Strength Index (RSI) as an affine representation of that coordinate. With a recursive mean M_t , signed increment smoother S_t , and absolute-increment smoother A_t , all using gain α , the matched finite-path identity is

$$X_t - M_t = \frac{1 - \alpha}{\alpha} S_t, \quad |S_t| \leq A_t.$$

For Wilder gain $\alpha = 1/n$, the normalized endpoint displacement is therefore bounded by $n - 1$. Writing $F_t = S_t/A_t$ gives the exact chain

$$\frac{RSI_t}{100} = \frac{1 + F_t}{2}, \quad L_t = \text{logit}(RSI_t/100) = 2 \operatorname{atanh}(F_t).$$

These identities are distribution-free. The exponential kernel has exact squared energy $1/(2n - 1)$ and exact Kish effective length $2n - 1$. Under stationary centered i.i.d. increments with variance $\sigma^2 < \infty$ and $m_1 = \mathbb{E}|\varepsilon| > 0$, the full log-odds coordinate satisfies

$$\frac{\sqrt{2n - 1} m_1}{2\sigma} L_t^{(n)} \Rightarrow N(0, 1).$$

Equivalently, $\sqrt{n} L_t^{(n)} / C_F \Rightarrow N(0, 1)$ with $C_F = \sqrt{2} \sigma / m_1$. Under a finite third absolute moment, an exact event reduction yields a Berry–Esseen bound of order $n^{-1/2}$ for the full nonlinear log ratio. Under stronger symmetry and light-tail assumptions, its variance admits the finite-horizon expansion

$$\operatorname{Var}(L_n) = v_1 \alpha + v_2 \alpha^2 + v_3 \alpha^3 + v_4 \alpha^4 + O(\alpha^5), \quad \alpha = 1/n,$$

where $v_1, \dots, v_4$ are explicit functions of normalized absolute moments. Series inversion yields a law-specific variance-normalizing length

$$g_F^*(n) = \frac{C_F^2}{\operatorname{Var}(L_n)} = n - K_F + \frac{A_F}{n} + \frac{B_F}{n^2} + O(n^{-3}).$$

For Laplace increments this gives $g_L^*(n) = n - 4/3 + 5/(18n) + 5/(27n^2) + O(n^{-3})$; Gaussian increments produce different, π -dependent coefficients. A companion Lean 4 development checks the exact pathwise identities, the Wilder recursion bridge, kernel energy, local log-odds transfer, and a finite-variance limit theorem for normalized finite linear rows. The stationary nonlinear limit, Berry–Esseen bound, and fourth-order analytic remainder are proved by conventional arguments; the coefficient algebra is independently reproduced by a pinned SymPy calculation. No trading rule or predictive claim is made.

Keywords: endpoint coordinate, RSI, log odds, Wilder smoothing, effective sample size, finite-horizon normalization, asymptotic expansion, Berry–Esseen, Lean 4.

1 Purpose and scope

Wilder’s RSI is conventionally introduced as a bounded oscillator [7]. The starting point here is different: the bounded scale is treated as a representation of two exponentially smoothed directional masses. Once that representation is written explicitly, a natural unbounded coordinate is the log ratio of those masses.

A prior practitioner paper proposed an empirical cross-lookback scaling and explicitly left its formal derivation as an open problem [4]. The present paper addresses only that mathematical problem. It does not attempt to justify any particular market threshold, adaptive zone, or trading interpretation.

Input scale and market interpretation

The process X is the input series to which Wilder smoothing is applied. The deterministic identities below are pathwise and therefore apply whether X is a price, a log price, or another real-valued series. The probabilistic results have a narrower interpretation: their assumptions concern the increments $\varepsilon_t = X_t - X_{t-1}$ of the chosen input. If $X_t = \log P_t$, then ε_t is a log return and the Gaussian and Laplace formulas describe RSI computed from log prices. If $X_t = P_t$, they instead describe an additive price-increment benchmark.

This distinction matters under the multiplicative price model

$$P_t = P_{t-1}e^{r_t}, \quad \Delta P_t = P_{t-1}(e^{r_t} - 1). \quad (1)$$

Even when the log returns r_t are i.i.d. and symmetric, the arithmetic price increments ΔP_t are state dependent and generally asymmetric. Thus the distribution-specific finite- n coefficients derived here are not exact theorems for classical price RSI under that model. A separate numerical supplement, `supplement/price_path_normality_audit/`, measures the small-return robustness of the same normalizers after symmetric log returns are mapped into positive price paths. No geometric-Brownian-motion theorem or empirical market model is claimed in this paper.

The logical and formal-verification status of the main results is summarized in Table 1. “Lean-checked” means that the corresponding theorem is accepted by the Lean kernel under assumptions visible in its type. It does not mean that those assumptions have been established for financial data.

Table 1: Status and assumptions of the principal results.

Result	Mathematical status	Formal status	Main assumptions
Endpoint, classical Wilder RSI, and log-odds identities	exact	Lean-checked	matched gain and initialization; positivity for finite log odds
Kernel energy and effective length	exact	Lean-checked	$n > 1$
Finite weighted linear-row CLT	theorem	Lean-checked	i.i.d., centered, finite variance; vanishing omitted squared-weight mass
Stationary nonlinear log-odds limit	theorem	conventional proof; Lean-checked components	i.i.d., centered, finite variance; stationary infinite past
Berry–Esseen approximation	theorem	conventional proof	above plus $\mathbb{E} \varepsilon ^3 < \infty$
Variance expansion through α^4	theorem	conventional proof	symmetry, no atom at zero, exponential and logarithmic moments
Laplace/Gaussian finite- n coefficients	corollaries	symbolically reproduced; not formalized as a Lean variance theorem	corresponding parametric law
No universal constant shift $n - c$	corollary	conventional proof	two covered laws with distinct K_F

2 Directional masses and the exact log-odds coordinate

Let

$$X_t = X_{t-1} + \varepsilon_t, \quad (2)$$

and for integer $n \geq 2$ define

$$\alpha_n = \frac{1}{n}, \quad \lambda_n = 1 - \frac{1}{n}. \quad (3)$$

Write positive and negative increment parts as

$$g_t = \max(\varepsilon_t, 0), \quad \ell_t = \max(-\varepsilon_t, 0), \quad (4)$$

and define Wilder smoothers

$$U_t = \lambda_n U_{t-1} + \alpha_n g_t, \quad D_t = \lambda_n D_{t-1} + \alpha_n \ell_t. \quad (5)$$

Set

$$A_t = U_t + D_t, \quad S_t = U_t - D_t, \quad (6)$$

and, whenever $A_t > 0$,

$$F_t = \frac{S_t}{A_t} = \frac{U_t - D_t}{U_t + D_t} \in [-1, 1]. \quad (7)$$

Then classical RSI may be written as

$$\frac{RSI_t}{100} = \frac{U_t}{U_t + D_t} = \frac{1 + F_t}{2}. \quad (8)$$

The equalities $S_t = U_t - D_t$ and $A_t = U_t + D_t$ are preserved step by step by the same Wilder recursion. Thus the directional masses used here are the classically smoothed up/down increments themselves, not masses introduced only by algebraically reconstructing a given pair (S_t, A_t) .

Proposition 1 (Exact log-odds identity). *If $U_t > 0$ and $D_t > 0$, then*

$$L_t := \log \frac{U_t}{D_t} = \log \frac{1 + F_t}{1 - F_t} = 2 \operatorname{atanh}(F_t) = \operatorname{logit}\left(\frac{RSI_t}{100}\right). \quad (9)$$

Proof. From $F = (U - D)/(U + D)$,

$$1 + F = \frac{2U}{U + D}, \quad 1 - F = \frac{2D}{U + D}.$$

Their ratio is U/D . The remaining identities are definitions of atanh and logit . $\square$

At $F = \pm 1$, L takes the extended-real values $\pm\infty$. No distributional statement follows from Proposition 1 alone.

3 Exact endpoint geometry

Define the recursive mean using the same Wilder gain,

$$M_t = \lambda_n M_{t-1} + \alpha_n X_t. \quad (10)$$

The signed smoother obeys

$$S_t = \lambda_n S_{t-1} + \alpha_n \varepsilon_t. \quad (11)$$

Proposition 2 (Endpoint displacement identity). *If the initialization satisfies*

$$X_0 - M_0 = (n - 1)S_0, \quad (12)$$

then for every $t \geq 0$,

$$\boxed{X_t - M_t = (n - 1)S_t.} \quad (13)$$

Proof. Since $X_t = X_{t-1} + \varepsilon_t$,

$$\begin{aligned} X_t - M_t &= X_{t-1} + \varepsilon_t - \lambda_n M_{t-1} - \alpha_n X_t \\ &= \lambda_n (X_{t-1} - M_{t-1}) + \lambda_n \varepsilon_t. \end{aligned}$$

Using the induction hypothesis and $(n-1)\alpha_n = \lambda_n$,

$$X_t - M_t = (n-1)(\lambda_n S_{t-1} + \alpha_n \varepsilon_t) = (n-1)S_t.$$

□

Consequently, with

$$B_{1,t} = \frac{X_t - M_t}{A_t}, \quad (14)$$

we have

$$B_{1,t} = (n-1)F_t \quad (15)$$

and therefore

$$\boxed{|B_{1,t}| \leq n-1.} \quad (16)$$

This is a deterministic support bound arising from first-absolute-moment geometry.

For arbitrary initialization, define

$$E_t = (X_t - M_t) - (n-1)S_t. \quad (17)$$

Direct substitution gives

$$E_t = \lambda_n E_{t-1} = \lambda_n^t E_0, \quad (18)$$

so the identity error decays geometrically.

4 Exact exponential-kernel geometry

For a generic gain $\alpha \in (0, 1)$, define the infinite exponential kernel

$$w_j = \alpha(1-\alpha)^j, \quad j \geq 0. \quad (19)$$

Proposition 3 (Kernel mass, energy, and effective length). *The kernel satisfies*

$$\sum_{j \geq 0} w_j = 1, \quad \sum_{j \geq 0} w_j^2 = \frac{\alpha}{2-\alpha}. \quad (20)$$

Consequently its Kish effective length is

$$N_{\text{eff}}(\alpha) := \frac{1}{\sum_{j \geq 0} w_j^2} = \frac{2-\alpha}{\alpha}. \quad (21)$$

For Wilder gain $\alpha = 1/n$,

$$\boxed{\sum_{j \geq 0} w_{n,j}^2 = \frac{1}{2n-1}, \quad N_{\text{eff}}(n) = 2n-1.} \quad (22)$$

Proof. Both statements are geometric-series identities:

$$\sum_{j \geq 0} \alpha(1-\alpha)^j = \frac{\alpha}{1-(1-\alpha)} = 1, \quad (23)$$

$$\sum_{j \geq 0} \alpha^2(1-\alpha)^{2j} = \frac{\alpha^2}{1-(1-\alpha)^2} = \frac{\alpha}{2-\alpha}. \quad (24)$$

The effective-length formulas follow by inversion. □

If ε_j are independent, centered, and have variance σ^2 , then the corresponding linear signed smoother has exact variance

$$\text{Var}\left(\sum_{j \geq 0} w_{n,j} \varepsilon_j\right) = \frac{\sigma^2}{2n-1}. \quad (25)$$

The number $2n-1$ is therefore the exact inverse-energy length of the *linear* exponential smoother. It is not an exact finite- n variance normalizer for the nonlinear ratio $F = S/A$ or for $L = 2 \operatorname{atanh}(F)$. Randomness of the denominator and curvature of the log-odds map generate the law-dependent corrections studied later.

5 Stationary infinite-past construction

Let $\{\varepsilon_t\}_{t \in \mathbb{Z}}$ be i.i.d. with

$$\mathbb{E}\varepsilon_t = 0, \quad 0 < \sigma^2 = \mathbb{E}\varepsilon_t^2 < \infty, \quad m_1 = \mathbb{E}|\varepsilon_t| > 0. \quad (26)$$

For each n , define stationary smoothers

$$S_t^{(n)} = \sum_{j=0}^{\infty} w_{n,j} \varepsilon_{t-j}, \quad (27)$$

$$A_t^{(n)} = \sum_{j=0}^{\infty} w_{n,j} |\varepsilon_{t-j}|, \quad (28)$$

where

$$w_{n,j} = \alpha_n \lambda_n^j. \quad (29)$$

Then

$$F_t^{(n)} = \frac{S_t^{(n)}}{A_t^{(n)}}, \quad L_t^{(n)} = 2 \operatorname{atanh}(F_t^{(n)}). \quad (30)$$

For any nondegenerate zero-mean law, positive and negative increments both occur with positive probability; under the infinite-past construction $U_t, D_t > 0$ almost surely.

6 Large-lookback limit

Theorem 1 (Asymptotic log-odds coordinate). *Under the stationary i.i.d. assumptions above,*

$$\boxed{\frac{\sqrt{2n-1} m_1}{2\sigma} L_t^{(n)} \Rightarrow N(0, 1)}. \quad (31)$$

Equivalently, on the conventional $\sqrt{n}$ scale, define

$$\boxed{C_F = \frac{\sqrt{2} \sigma}{m_1} = \frac{\sqrt{2} \sigma}{\mathbb{E}|\varepsilon|}}, \quad (32)$$

so that

$$\frac{\sqrt{n}}{C_F} L_t^{(n)} \Rightarrow N(0, 1). \quad (33)$$

Proof. The exact squared-weight sum is

$$\sum_{j \geq 0} w_{n,j}^2 = \frac{1}{2n-1}. \quad (34)$$

Hence

$$\text{Var}(S_t^{(n)}) = \frac{\sigma^2}{2n-1}. \quad (35)$$

To justify the triangular-array step without appealing directly to an infinite-row theorem, put

$$J_n = \lceil n \log n \rceil, \quad B_n = \sum_{j=0}^{J_n} w_{n,j}^2, \quad b_{n,j} = \frac{w_{n,j}}{\sqrt{B_n}} \quad (0 \leq j \leq J_n). \quad (36)$$

Then $\sum_{j=0}^{J_n} b_{n,j}^2 = 1$ and

$$\max_{0 \leq j \leq J_n} |b_{n,j}| \leq \frac{\sqrt{2n-1}}{n\sqrt{1-\lambda_n^{2(J_n+1)}}} \rightarrow 0. \quad (37)$$

For every fixed $\eta > 0$, independence and identical distribution give the Lindeberg bound

$$\sum_{j=0}^{J_n} \mathbb{E} \left[\left(\frac{b_{n,j} \varepsilon_{t-j}}{\sigma} \right)^2 \mathbf{1}_{\{|b_{n,j} \varepsilon_{t-j}| > \eta \sigma\}} \right] \quad (38)$$

$$\leq \frac{1}{\sigma^2} \mathbb{E} \left[\varepsilon^2 \mathbf{1}_{\{|\varepsilon| > \eta \sigma / \max_j |b_{n,j}|\}} \right] \rightarrow 0, \quad (39)$$

because $\mathbb{E} \varepsilon^2 < \infty$. The finite-row Lindeberg–Feller theorem therefore applies. The omitted tail is negligible on the full standard deviation scale, since

$$\frac{\sum_{j > J_n} w_{n,j}^2}{\sum_{j \geq 0} w_{n,j}^2} = \lambda_n^{2(J_n+1)} \leq \exp\left(-\frac{2(J_n+1)}{n}\right) \leq n^{-2}. \quad (40)$$

Also $B_n / \sum_{j \geq 0} w_{n,j}^2 \rightarrow 1$. Lindeberg–Feller and Slutsky hence yield

$$\frac{\sqrt{2n-1}}{\sigma} S_t^{(n)} \Rightarrow N(0, 1). \quad (41)$$

Also

$$\mathbb{E} A_t^{(n)} = m_1, \quad \text{Var}(A_t^{(n)}) = \frac{\text{Var}(|\varepsilon|)}{2n-1} \rightarrow 0, \quad (42)$$

so $A_t^{(n)} \rightarrow_p m_1$. Slutsky gives

$$\frac{\sqrt{2n-1} m_1}{\sigma} F_t^{(n)} \Rightarrow N(0, 1). \quad (43)$$

Thus $F_t^{(n)} = O_p(n^{-1/2})$. On every fixed compact subset of $(-1, 1)$,

$$2 \operatorname{atanh} x = 2x + O(x^3). \quad (44)$$

Moreover, for fixed $c \in (0, 1)$, Chebyshev’s inequality gives the explicit split

$$\mathbb{P}(|F_t^{(n)}| > c) \leq \mathbb{P}\left(A_t^{(n)} < \frac{m_1}{2}\right) + \mathbb{P}\left(|S_t^{(n)}| > \frac{cm_1}{2}\right) \quad (45)$$

$$\leq \frac{4 \operatorname{Var}(|\varepsilon|)}{m_1^2(2n-1)} + \frac{4\sigma^2}{c^2 m_1^2(2n-1)} = O(n^{-1}). \quad (46)$$

On $|F_t^{(n)}| \leq c$ there is a finite C_c such that $|2 \operatorname{atanh} F_t^{(n)} - 2F_t^{(n)}| \leq C_c |F_t^{(n)}|^3$. Since $\sqrt{n} F_t^{(n)} = O_p(1)$, the right-hand side multiplied by $\sqrt{n}$ converges to zero in probability; the complement has probability $O(n^{-1})$. Therefore

$$\sqrt{2n-1} [L_t^{(n)} - 2F_t^{(n)}] \rightarrow_p 0, \quad (47)$$

and the effective-length form follows. Finally,

$$\frac{\sqrt{n}}{C_F} = \frac{\sqrt{2n-1} m_1}{2\sigma} \sqrt{\frac{2n}{2n-1}},$$

and the last factor tends to one, proving the equivalent $\sqrt{n}$ form. $\square$

Remark 1. *The theorem is a distributional limit. Under only a finite second moment it does not, by itself, assert convergence of the variances of the normalized log-odds variables; such a conclusion would require an additional uniform integrability argument.*

7 A quantitative Gaussian approximation

The preceding proof uses a ratio approximation. A sharper finite- n statement can be obtained from the Berry–Esseen inequality [1, 3] without approximating the random denominator.

Theorem 2 (Berry–Esseen bound for the full log ratio). *Assume in addition that*

$$\rho_3 := \mathbb{E}|\varepsilon|^3 < \infty. \quad (48)$$

Then there exist finite constants K, N , depending only on the innovation law, such that for all $n \geq N$,

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\sqrt{n}}{C_F} L_t^{(n)} \leq x \right) - \Phi(x) \right| \leq \frac{K}{\sqrt{n}}. \quad (49)$$

Proof. For $h \in \mathbb{R}$, define

$$\xi(h) = g - e^h \ell, \quad (50)$$

where $g = \max(\varepsilon, 0)$ and $\ell = \max(-\varepsilon, 0)$. Then

$$T_n(h) := U_t - e^h D_t = \sum_{j \geq 0} w_{n,j} \xi_{t-j}(h). \quad (51)$$

Since $U_t, D_t > 0$ almost surely,

$$\{L_t^{(n)} \leq h\} = \{T_n(h) \leq 0\}. \quad (52)$$

This reduction is exact.

Because $\mathbb{E}\varepsilon = 0$,

$$\mathbb{E}g = \mathbb{E}\ell = \frac{m_1}{2}, \quad (53)$$

so

$$\mu(h) := \mathbb{E}\xi(h) = \frac{m_1}{2}(1 - e^h). \quad (54)$$

Let

$$v(h) := \text{Var}(\xi(h)). \quad (55)$$

Since $g\ell = 0$ pointwise,

$$v(h) = \mathbb{E}g^2 + e^{2h}\mathbb{E}\ell^2 - \mu(h)^2. \quad (56)$$

Thus v is analytic in h , $v(0) = \sigma^2 > 0$, and both v and $v^{-1/2}$ are continuously differentiable on a sufficiently small neighborhood of zero. Choose $h_0 > 0$ small enough that

$$v_* := \inf_{|h| \leq h_0} v(h) > 0. \quad (57)$$

The third absolute moment is uniformly controlled on the same window. Indeed, $|\xi(h)| \leq e^{h_0}|\varepsilon|$ for $|h| \leq h_0$, Jensen's inequality gives $|\mu(h)|^3 \leq \mathbb{E}|\xi(h)|^3$, and $|a - b|^3 \leq 4(|a|^3 + |b|^3)$, hence

$$\sup_{|h| \leq h_0} \mathbb{E}|\xi(h) - \mu(h)|^3 \leq 8e^{3h_0}\rho_3. \quad (58)$$

The exact weight sums are

$$\sum_{j \geq 0} w_j^2 = \frac{1}{2n-1}, \quad \sum_{j \geq 0} w_j^3 = \frac{1}{3n^2-3n+1}. \quad (59)$$

Consequently the Berry–Esseen Lyapunov ratio is uniformly bounded by

$$\frac{\mathbb{E}|\xi(h) - \mu(h)|^3 \sum_j w_j^3}{v(h)^{3/2} (\sum_j w_j^2)^{3/2}} \leq \frac{8e^{3h_0}\rho_3}{v_*^{3/2}} \frac{(2n-1)^{3/2}}{3n^2-3n+1} \quad (60)$$

$$\leq \frac{K_{BE}}{\sqrt{n}}, \quad |h| \leq h_0, \quad (61)$$

for a finite law-dependent constant K_{BE} .

For completeness, the passage from finite to infinite weighted sums can be made quantitative rather than implicit. Put

$$J_n = \lceil n \log n \rceil, \quad T_{n,J}(h) = \sum_{j=0}^{J_n} w_j \xi_{t-j}(h), \quad R_{n,J}(h) = T_n(h) - T_{n,J}(h). \quad (62)$$

Since $\lambda_n = 1 - 1/n$,

$$\sum_{j>J_n} w_j^2 = \frac{1}{2n-1} \lambda_n^{2(J_n+1)} = O(n^{-3}), \quad (63)$$

$$\sum_{j>J_n} w_j^3 = \frac{1}{3n^2-3n+1} \lambda_n^{3(J_n+1)} = O(n^{-5}). \quad (64)$$

Therefore, uniformly for $|h| \leq h_0$,

$$\text{Var}(R_{n,J}(h)) = O(n^{-3}), \quad (65)$$

whereas $\text{Var}(T_n(h) - \mu(h)) = v(h)/(2n-1) \asymp n^{-1}$. Thus the omitted tail has normalized variance $O(n^{-2})$.

Write

$$\sigma_{n,J}^2 = v(h) \sum_{j=0}^{J_n} w_j^2, \quad \sigma_n^2 = \frac{v(h)}{2n-1}. \quad (66)$$

Then

$$\frac{\sigma_{n,J}^2}{\sigma_n^2} = 1 - \lambda_n^{2(J_n+1)} = 1 + O(n^{-2}), \quad \frac{\sigma_{n,J}}{\sigma_n} = 1 + O(n^{-2}), \quad (67)$$

uniformly for $|h| \leq h_0$. Apply the Berry–Esseen inequality for independent, not necessarily identically distributed, centered summands (Petrov [5]) to $T_{n,J}(h)$ with its natural scale $\sigma_{n,J}$. The truncated second- and third-weight sums differ from their infinite counterparts by relative $O(n^{-2})$ and $O(n^{-3})$, respectively, so their Lyapunov ratios differ by a factor $1 + O(n^{-2})$. Thus the same $K_{BE}/\sqrt{n}$ bound holds for the truncated statistic after enlarging the constant. Replacing $\sigma_{n,J}$ by σ_n adds only $O(n^{-2})$ to the normal comparison, because $\sup_z |\Phi(cz) - \Phi(z)| = O(|c-1|)$ as $c \rightarrow 1$. To transfer the bound to $T_n(h)$, use, for any $\eta > 0$, the elementary smoothing inequality

$$\sup_z \left| \mathbb{P}\left(\frac{S+R}{\sigma_n} \leq z\right) - \mathbb{P}\left(\frac{S}{\sigma_n} \leq z\right) \right| \leq \mathbb{P}(|R| > \eta \sigma_n) + \sup_z \mathbb{P}(z - \eta < S/\sigma_n \leq z + \eta), \quad (68)$$

Taking $S = T_{n,J}(h) - \mathbb{E}T_{n,J}(h)$, $R = R_{n,J}(h) - \mathbb{E}R_{n,J}(h)$, and $\eta = n^{-1/2}$, Chebyshev's inequality and the normalized tail variance $O(n^{-2})$ give an $O(n^{-1})$ first term; the second is $O(n^{-1/2})$ by the Gaussian interval bound plus the finite-sum Berry–Esseen error. Hence, uniformly for $|h| \leq h_0$,

$$\boxed{\sup_z \left| \mathbb{P}\left(\frac{T_n(h) - \mu(h)}{\sqrt{v(h)/(2n-1)}} \leq z\right) - \Phi(z) \right| \leq \frac{K_1}{\sqrt{n}}.} \quad (69)$$

Now set

$$h_n(x) = \frac{C_F x}{\sqrt{n}} \quad (70)$$

and define

$$z_n(h) = -\mu(h) \frac{\sqrt{2n-1}}{\sqrt{v(h)}}. \quad (71)$$

The exact event reduction gives

$$\mathbb{P}\left(\frac{\sqrt{n}}{C_F} L_t^{(n)} \leq x\right) = \mathbb{P}\left(\frac{T_n(h_n(x)) - \mu(h_n(x))}{\sqrt{v(h_n(x))/(2n-1)}} \leq z_n(h_n(x))\right). \quad (72)$$

For $x \neq 0$, put $h = h_n(x)$. Since $x = h\sqrt{n}/C_F$ and $C_F = \sqrt{2}\sigma/m_1$, direct substitution gives the exact ratio

$$r_n(h) := \frac{z_n(h)}{x} = \sqrt{\frac{2n-1}{2n}} \frac{\sigma}{\sqrt{v(h)}} \frac{e^h - 1}{h}. \quad (73)$$

Define

$$q(h) = \frac{\sigma}{\sqrt{v(h)}} \frac{e^h - 1}{h}, \quad q(0) = 1. \quad (74)$$

After reducing h_0 if necessary, q is continuously differentiable on $[-h_0, h_0]$. Let

$$M = \sup_{|h| \leq h_0} |q'(h)| < \infty. \quad (75)$$

Using

$$0 \leq 1 - \sqrt{1 - \frac{1}{2n}} \leq \frac{1}{2n}, \quad (76)$$

we obtain, uniformly for $|h| \leq h_0$,

$$|r_n(h) - 1| \leq M|h| + \frac{1}{2n}. \quad (77)$$

Choose h_0 smaller, if needed, and then N larger, so that for all $n \geq N$ and $|h| \leq h_0$,

$$\frac{1}{2} \leq r_n(h) \leq \frac{3}{2}. \quad (78)$$

Thus $z_n(h_n(x))$ and x have the same sign and

$$|z_n(h_n(x))| \geq \frac{|x|}{2} \quad (79)$$

throughout the central range $|h_n(x)| \leq h_0$. Moreover,

$$|z_n(h_n(x)) - x| \leq MC_F \frac{x^2}{\sqrt{n}} + \frac{|x|}{2n}. \quad (80)$$

The mean-value theorem now gives, for some point between x and $z_n(h_n(x))$,

$$|\Phi(z_n(h_n(x))) - \Phi(x)| \leq |z_n(h_n(x)) - x| \varphi(|x|/2) \quad (81)$$

$$\leq \left(MC_F \frac{x^2}{\sqrt{n}} + \frac{|x|}{2n} \right) \varphi(|x|/2). \quad (82)$$

Because

$$\sup_{x \in \mathbb{R}} (1 + x^2) e^{-x^2/8} < \infty, \quad (83)$$

there is a finite K_2 such that

$$\sup_{|h_n(x)| \leq h_0} |\Phi(z_n(h_n(x))) - \Phi(x)| \leq \frac{K_2}{\sqrt{n}}. \quad (84)$$

Combining this with the Berry–Esseen bound gives the desired $O(n^{-1/2})$ approximation on the central range.

For completeness, let

$$x_n^* = \frac{h_0 \sqrt{n}}{C_F}. \quad (85)$$

If $x > x_n^*$, monotonicity implies

$$1 - F_n(x) \leq 1 - F_n(x_n^*) \leq 1 - \Phi(x_n^*) + \frac{K_1 + K_2}{\sqrt{n}}, \quad (86)$$

where F_n is the distribution function of $\sqrt{n}L_t^{(n)}/C_F$. Also $1 - \Phi(x) \leq 1 - \Phi(x_n^*)$. Since $x_n^* = (h_0/C_F)\sqrt{n}$, the Gaussian tail is $O(e^{-cn})$. The analogous argument applies for $x < -x_n^*$. Absorbing this exponentially small boundary term into the constant proves the theorem globally. $\square$

Corollary 1 (Gaussian and Laplace scale constants). *For Gaussian increments,*

$$\boxed{C_G = \sqrt{\pi}.} \quad (87)$$

For centered Laplace increments,

$$\boxed{C_L = 2.} \quad (88)$$

8 Symmetric finite-horizon variance expansion

This section is deliberately narrower than Theorems 1–2. Symmetry is essential to the displayed coefficient formulas. In particular, the expansion below is *not* an asymmetric-law analogue of the earlier Gaussian limit.

Assume henceforth that ε is symmetric, $\mathbb{P}(\varepsilon = 0) = 0$, and define

$$R = \frac{|\varepsilon|}{m_1}, \quad m_k = \mathbb{E}R^k, \quad m_1 = 1. \quad (89)$$

Symmetry implies the representation

$$\frac{\varepsilon}{m_1} = QR, \quad (90)$$

where Q is a Rademacher sign independent of R .

For a fixed n , abbreviate $\alpha = 1/n$, $\lambda = 1 - \alpha$, and $w_j = \alpha\lambda^j$. Define centered weighted sums

$$x = \sum_{j \geq 0} w_j Q_j R_j, \quad y = \sum_{j \geq 0} w_j (R_j - 1). \quad (91)$$

Then

$$\frac{S}{m_1} = x, \quad \frac{A}{m_1} = 1 + y, \quad (92)$$

so

$$F = \frac{x}{1 + y}, \quad L = 2 \operatorname{atanh}\left(\frac{x}{1 + y}\right). \quad (93)$$

The distribution of L is symmetric, hence $\mathbb{E}L = 0$ whenever the expectation exists.

8.1 Weight sums and joint cumulants

For integer $r \geq 2$, let

$$s_r(\alpha) = \sum_{j \geq 0} w_j^r = \frac{\alpha^r}{1 - (1 - \alpha)^r}. \quad (94)$$

Through order α^4 ,

$$s_2 = \frac{\alpha}{2} + \frac{\alpha^2}{4} + \frac{\alpha^3}{8} + \frac{\alpha^4}{16} + O(\alpha^5), \quad (95)$$

$$s_3 = \frac{\alpha^2}{3} + \frac{\alpha^3}{3} + \frac{2\alpha^4}{9} + O(\alpha^5), \quad (96)$$

$$s_4 = \frac{\alpha^3}{4} + \frac{3\alpha^4}{8} + O(\alpha^5), \quad (97)$$

$$s_5 = \frac{\alpha^4}{5} + O(\alpha^5), \quad (98)$$

$$s_r = O(\alpha^5), \quad r \geq 6. \quad (99)$$

Let $q = QR$ and $r = R - 1$. For $a, b \geq 0$, define the one-observation joint cumulant

$$c_{a,b} = \operatorname{cum}(\underbrace{q, \dots, q}_a, \underbrace{r, \dots, r}_b). \quad (100)$$

Table 2: One-observation joint cumulants required through $O(\alpha^4)$.

(a, b)	$c_{a,b}$
$(2, 0)$	m_2
$(0, 2)$	$m_2 - 1$
$(2, 1)$	$m_3 - m_2$
$(0, 3)$	$m_3 - 3m_2 + 2$
$(4, 0)$	$m_4 - 3m_2^2$
$(2, 2)$	$m_4 - 2m_3 + 2m_2 - m_2^2$
$(0, 4)$	$m_4 - 4m_3 + 12m_2 - 3m_2^2 - 6$
$(4, 1)$	$m_5 - m_4 - 6m_2m_3 + 6m_2^2$
$(2, 3)$	$m_5 - 3m_4 + 6m_3 - 6m_2 + 6m_2^2 - 4m_2m_3$
$(0, 5)$	$m_5 - 5m_4 + 20m_3 - 60m_2 + 30m_2^2 - 10m_2m_3 + 24$

Every $c_{a,b}$ with odd a vanishes. The nonzero cumulants needed through total order five are listed in Table 2.

Because the observations are independent across j , cumulants add, giving the exact identity

$$\boxed{\text{cum}(\underbrace{x, \dots, x}_a, \underbrace{y, \dots, y}_b) = s_{a+b}(\alpha)c_{a,b}.} \quad (101)$$

8.2 Taylor polynomial for the squared log odds

Set

$$H(x, y) = \left(\frac{L}{2}\right)^2 = \text{atanh}^2\left(\frac{x}{1+y}\right). \quad (102)$$

Using

$$\text{atanh}^2 z = z^2 + \frac{2}{3}z^4 + \frac{23}{45}z^6 + \frac{44}{105}z^8 + O(z^{10}), \quad (103)$$

and expanding $(1+y)^{-2k}$, the terms of total degree at most eight are

$$P_8(x, y) = x^2 - 2x^2y + 3x^2y^2 - 4x^2y^3 + 5x^2y^4 - 6x^2y^5 + 7x^2y^6 \quad (104)$$

$$+ \frac{2}{3}x^4 - \frac{8}{3}x^4y + \frac{20}{3}x^4y^2 - \frac{40}{3}x^4y^3 + \frac{70}{3}x^4y^4 \quad (105)$$

$$+ \frac{23}{45}x^6 - \frac{46}{15}x^6y + \frac{161}{15}x^6y^2 + \frac{44}{105}x^8. \quad (106)$$

The moment–cumulant formula applied to the cumulants above determines $\mathbb{E}P_8$ exactly through α^4 . The next lemma records the order bookkeeping used both in the coefficient calculation and in the remainder proof.

Lemma 1 (Weighted centered-moment order). *Suppose the required moments exist. For integers $a, b \geq 0$ with $d = a + b \geq 2$,*

$$\mathbb{E}[x^a y^b] = O\left(\alpha^{\lceil d/2 \rceil}\right) \quad (107)$$

whenever the moment–cumulant expansion is taken to a fixed finite order as $\alpha \downarrow 0$.

Proof. Both x and y are centered, so singleton cumulants vanish. In the moment–cumulant formula, a partition π of d positions therefore contains no singleton blocks. A block of size r contributes a joint cumulant of order $s_r(\alpha) = O(\alpha^{r-1})$. Thus the product associated with π has order

$$\alpha^{\sum_{B \in \pi} (|B|-1)} = \alpha^{d-|\pi|}. \quad (108)$$

Since $|\pi| \leq \lfloor d/2 \rfloor$, the exponent is at least $\lceil d/2 \rceil$. $\square$

8.3 Remainder control

The singularity of atanh at ± 1 prevents an unconditional global Taylor argument. The next lemma localizes the statistic and shows that the complement is negligible at every polynomial order needed below.

Lemma 2 (Localization and Taylor remainder). *Assume, in addition to symmetry and $\mathbb{P}(\varepsilon = 0) = 0$, that for $R = |\varepsilon|/m_1$ there exists $\theta > 0$ such that*

$$\mathbb{E}e^{\theta R} < \infty \quad (109)$$

and

$$\mathbb{E}|\log R|^4 < \infty. \quad (110)$$

Then, for some fixed $\delta \in (0, 1/3)$ and constants $c, C > 0$,

$$\mathbb{P}(|x| > \delta \text{ or } |y| > \delta) \leq Ce^{-c/\alpha}. \quad (111)$$

Moreover,

$$\mathbb{E}[H(x, y) - P_8(x, y)] = O(\alpha^5). \quad (112)$$

Proof. We first make the concentration step explicit. If a centered random variable Z has a moment generating function in a neighborhood of zero, then there exist finite constants $\nu, b > 0$ such that

$$\mathbb{E}e^{tZ} \leq \exp\left(\frac{\nu^2 t^2}{2}\right), \quad |t| \leq b^{-1}. \quad (113)$$

For a finite deterministic weight sequence $a_0, \dots, a_J$, independence therefore gives

$$\mathbb{E} \exp\left(t \sum_{j=0}^J a_j Z_j\right) \leq \exp\left(\frac{\nu^2 t^2}{2} \sum_{j=0}^J a_j^2\right), \quad |t| \leq \frac{1}{b \max_{0 \leq j \leq J} |a_j|}. \quad (114)$$

The Chernoff argument, optimized over the permitted range of t , yields the weighted Bernstein bound (a standard sub-exponential concentration estimate; see, e.g., Boucheron, Lugosi, and Massart [2])

$$\mathbb{P}\left(\left|\sum_{j=0}^J a_j Z_j\right| \geq u\right) \leq 2 \exp\left[-\frac{1}{2} \min\left\{\frac{u^2}{\nu^2 \sum_{j=0}^J a_j^2}, \frac{u}{b \max_{0 \leq j \leq J} |a_j|}\right\}\right]. \quad (115)$$

For the geometric Wilder weights, $\sum_j w_j \mathbb{E}|Z_j| < \infty$, so the weighted series converges absolutely almost surely. Applying the finite bound to partial sums and then letting $J \rightarrow \infty$ therefore gives the same inequality with the full sums $\sum_j a_j^2$ and $\max_j |a_j|$. The assumption $\mathbb{E}e^{\theta R} < \infty$ implies such a local mgf bound for both centered variables $Z = QR$ and $Z = R - 1$. Indeed, for $|t| \leq \theta$ and Q independent Rademacher,

$$\mathbb{E}e^{tQR} = \frac{1}{2} (\mathbb{E}e^{tR} + \mathbb{E}e^{-tR}) \leq \frac{1}{2} (\mathbb{E}e^{|t|R} + 1) < \infty, \quad (116)$$

while

$$\mathbb{E}e^{t(R-1)} = e^{-t} \mathbb{E}e^{tR} \quad (117)$$

is finite for positive t in a neighborhood of zero, and for negative t it is bounded by $e^{|t|}$ because $R \geq 0$. Since both variables are centered, finiteness of the mgf on a two-sided neighborhood of zero implies the displayed local quadratic mgf bound after possibly shrinking that neighborhood. Taking $a_j = w_j$ gives

$$\sum_j w_j^2 = s_2(\alpha) = \frac{\alpha}{2 - \alpha} \leq \alpha, \quad \max_j w_j = \alpha. \quad (118)$$

Hence, for every fixed $\delta > 0$, possibly with different constants for x and y ,

$$\mathbb{P}(|x| > \delta) + \mathbb{P}(|y| > \delta) \leq 4 \exp\left(-\frac{c\delta}{\alpha}\right) \quad (119)$$

for all sufficiently small α . Renaming constants gives the stated $Ce^{-c/\alpha}$ bound. Let

$$G_\delta = \{|x| \leq \delta, |y| \leq \delta\} \quad (120)$$

with $\delta < 1/3$. On G_δ ,

$$\left| \frac{x}{1+y} \right| \leq \frac{\delta}{1-\delta} < 1, \quad (121)$$

so H is analytic on a compact neighborhood containing G_δ . Its multivariate Taylor expansion through total degree nine may be written as

$$H = P_8 + P_9 + R_{10}, \quad (122)$$

where P_9 is homogeneous of total degree nine and

$$|R_{10}(x, y)| \leq C_\delta(|x| + |y|)^{10} \quad (123)$$

on G_δ . Lemma 1 gives $\mathbb{E}P_9 = O(\alpha^5)$. The same bound holds after restriction to G_δ . More generally, if $P(x, y)$ is any fixed polynomial whose second moment is finite, then

$$\mathbb{E}[|P(x, y)|\mathbf{1}_{G_\delta^c}] \leq (\mathbb{E}P(x, y)^2)^{1/2} \mathbb{P}(G_\delta^c)^{1/2} = O(\alpha^M) \quad (124)$$

for every fixed $M > 0$: its moment factor grows at most polynomially in α^{-1} (and in fact remains bounded for the fixed-degree polynomials used here), whereas $\mathbb{P}(G_\delta^c)^{1/2} \leq Ce^{-c/(2\alpha)}$ dominates every power of α . Applying this observation to P_9 makes the restriction step explicit. By Rosenthal's inequality [6], for independent centered variables Z_j and $p = 10$,

$$\mathbb{E} \left| \sum_j a_j Z_j \right|^{10} \leq C_{10} \left[\left(\sum_j a_j^2 \mathbb{E} Z_j^2 \right)^5 + \sum_j |a_j|^{10} \mathbb{E} |Z_j|^{10} \right]. \quad (125)$$

This inequality is first applied to finite partial sums. Because $\sum_j |w_j|^{10} < \infty$, $\sum_j w_j^2 < \infty$, and the innovations have a finite tenth moment, those partial sums converge in L^{10} ; passage to the infinite sum therefore preserves the displayed bound. For $a_j = w_j$, the first term is $O(s_2^5) = O(\alpha^5)$, while $\sum_j w_j^{10} = s_{10}(\alpha) = O(\alpha^9)$; the exponential-moment assumption supplies the required finite tenth moments. Hence

$$\mathbb{E}|x|^{10} + \mathbb{E}|y|^{10} = O(\alpha^5). \quad (126)$$

so

$$\mathbb{E}[|R_{10}|\mathbf{1}_{G_\delta}] = O(\alpha^5). \quad (127)$$

It remains to control G_δ^c . Write the normalized positive directional mass as

$$\bar{U} = \frac{U}{m_1} = \sum_{j \geq 0} w_j \mathbf{1}_{\{Q_j = +1\}} R_j \quad (128)$$

and analogously $\bar{D}$. Let T_+ be the first $j \geq 0$ with $Q_j = +1$. Then T_+ is geometric with parameter $1/2$ and is independent of the corresponding magnitude R_{T_+} , while

$$\bar{U} \geq w_{T_+} R_{T_+}. \quad (129)$$

Hence

$$(-\log \bar{U})_+ \leq \log(1/\alpha) + T_+[-\log(1-\alpha)] + (-\log R_{T_+})_+. \quad (130)$$

Since T_+ has moments of every order, $-\log(1 - \alpha) = O(\alpha)$, and $\mathbb{E}|\log R|^4 < \infty$,

$$\mathbb{E}[(-\log \bar{U})_+^4] = O(\log^4(1/\alpha)). \quad (131)$$

For the positive logarithmic part, $\log^+ z \leq z$ and Jensen's inequality give a uniform fourth-moment bound. The same argument applies to $\bar{D}$. Therefore

$$\mathbb{E}|L|^4 = O(\log^4(1/\alpha)). \quad (132)$$

By Cauchy–Schwarz,

$$\mathbb{E}[L^2 \mathbf{1}_{G_\delta^c}] \leq (\mathbb{E}L^4)^{1/2} \mathbb{P}(G_\delta^c)^{1/2} = O\left(\log^2(1/\alpha) e^{-c/(2\alpha)}\right), \quad (133)$$

which is $O(\alpha^M)$ for every fixed M . In particular it is $O(\alpha^5)$. Since P_8 is a fixed polynomial and all polynomial moments of x and y are finite under the exponential-moment assumption, the same Cauchy–Schwarz argument gives $\mathbb{E}[P_8 \mathbf{1}_{G_\delta^c}] = O(\alpha^M)$ for every fixed M . Combining the localized Taylor bound and both complement bounds proves the result. $\square$

8.4 Variance coefficients

Theorem 3 (Finite-horizon variance expansion under symmetry). *Under the assumptions of Lemma 2,*

$$\boxed{\text{Var}(L_n) = v_1\alpha + v_2\alpha^2 + v_3\alpha^3 + v_4\alpha^4 + O(\alpha^5), \quad \alpha = 1/n,} \quad (134)$$

where

$$v_1 = 2m_2, \quad (135)$$

$$v_2 = \frac{15m_2^2 + 2m_2 - 8m_3}{3}, \quad (136)$$

$$v_3 = \frac{64m_2^3 + 5m_2^2 - 64m_2m_3 + m_2 - 2m_3 + 11m_4}{3}, \quad (137)$$

$$v_4 = \frac{1}{90} \left(11700m_2^4 - 17040m_2^2m_3 + 70m_2^2 + 224m_2m_3 \right. \quad (138)$$

$$\left. + 4020m_2m_4 + 18m_2 + 2400m_3^2 - 18m_3 - 99m_4 - 480m_5 \right). \quad (139)$$

Proof. Symmetry gives $\mathbb{E}L = 0$, so $\text{Var}(L) = \mathbb{E}L^2 = 4\mathbb{E}H(x, y)$. By Lemma 2,

$$\mathbb{E}H = \mathbb{E}P_8 + O(\alpha^5). \quad (140)$$

Every moment in $\mathbb{E}P_8$ is evaluated by the standard moment–cumulant formula using

$$\text{cum}(x^{[a]}, y^{[b]}) = s_{a+b}(\alpha) c_{a,b}, \quad (141)$$

where $x^{[a]}$ denotes a repeated arguments. Because $s_r = O(\alpha^{r-1})$, blocks of order $r \geq 6$ contribute only $O(\alpha^5)$ and may be omitted. Thus only the cumulants in Table 2 are needed. Substituting their exact values together with the expansions of s_2, s_3, s_4, s_5 into $\mathbb{E}P_8$, collecting powers of α , and multiplying by four gives the displayed coefficients.

The symbolic operation is finite: it consists only of the moment–cumulant partition formula applied to the monomials in P_8 . The accompanying rational-arithmetic script reproduces this calculation exactly; no numerical fitting or simulation enters the coefficient derivation. $\square$

Remark 2 (Direct check of the v_2 coefficient). *The second coefficient can be verified without the full partition calculation. Up to $O(\alpha^3)$,*

$$\left(\frac{L}{2}\right)^2 = x^2 - 2x^2y + 3x^2y^2 + \frac{2}{3}x^4 + O_{L^1}(\alpha^3). \quad (142)$$

The required moments are

$$\mathbb{E}x^2 = m_2 s_2, \quad (143)$$

$$\mathbb{E}x^2 y = (m_3 - m_2) s_3, \quad (144)$$

$$\mathbb{E}x^2 y^2 = m_2(m_2 - 1) s_2^2 + O(\alpha^3), \quad (145)$$

$$\mathbb{E}x^4 = 3m_2^2 s_2^2 + O(\alpha^3). \quad (146)$$

Using $s_2 = \alpha/2 + \alpha^2/4 + O(\alpha^3)$ and $s_3 = \alpha^2/3 + O(\alpha^3)$ gives

$$\text{Var}(L) = 2m_2\alpha + \frac{15m_2^2 + 2m_2 - 8m_3}{3}\alpha^2 + O(\alpha^3), \quad (147)$$

independently reproducing v_1 and v_2 .

9 Variance-normalizing length

For symmetric laws satisfying Theorem 3, define the exact variance-normalizing factor

$$\boxed{g_F^*(n) = \frac{C_F^2}{\text{Var}(L_n)}}. \quad (148)$$

Since $C_F^2 = 2m_2 = v_1$, the standardized coordinate

$$Z_n^* = \frac{\sqrt{g_F^*(n)}}{C_F} L_n \quad (149)$$

has variance exactly one by construction.

Theorem 3 yields an explicit asymptotic expansion for $g_F^*(n)$.

Corollary 2 (Series inversion). *Let*

$$K_F = \frac{v_2}{v_1}, \quad (150)$$

$$A_F = K_F^2 - \frac{v_3}{v_1}, \quad (151)$$

and

$$B_F = -K_F^3 + 2K_F \frac{v_3}{v_1} - \frac{v_4}{v_1}. \quad (152)$$

Then

$$\boxed{g_F^*(n) = n - K_F + \frac{A_F}{n} + \frac{B_F}{n^2} + O(n^{-3})}. \quad (153)$$

Proof. Write

$$\text{Var}(L_n) = v_1 \alpha \left[1 + \frac{v_2}{v_1} \alpha + \frac{v_3}{v_1} \alpha^2 + \frac{v_4}{v_1} \alpha^3 + O(\alpha^4) \right]. \quad (154)$$

Because $C_F^2 = v_1$,

$$g_F^*(n) = \frac{1}{\alpha} \frac{1}{1 + a\alpha + b\alpha^2 + c\alpha^3 + O(\alpha^4)}, \quad (155)$$

with $a = v_2/v_1$, $b = v_3/v_1$, $c = v_4/v_1$. Using

$$(1 + u)^{-1} = 1 - u + u^2 - u^3 + O(u^4) \quad (156)$$

and collecting powers gives

$$g_F^*(n) = \frac{1}{\alpha} - a + (a^2 - b)\alpha + (-a^3 + 2ab - c)\alpha^2 + O(\alpha^3), \quad (157)$$

which is the stated result after $\alpha = 1/n$. $\square$

Corollary 3 (No universal constant lookback shift). *Consider two symmetric light-tailed innovation laws F_1, F_2 satisfying Theorem 3 and having different first variance-shift coefficients $K_{F_1} \neq K_{F_2}$. For a law-independent constant c , define*

$$Z_{F,c}(n) = \frac{\sqrt{n-c}}{C_F} L_n. \quad (158)$$

Then

$$\text{Var}(Z_{F,c}(n)) = 1 + \frac{K_F - c}{n} + O(n^{-2}). \quad (159)$$

Consequently no single constant c can make

$$\text{Var}(Z_{F_i,c}(n)) = 1 + o(n^{-1}) \quad (160)$$

simultaneously for $i = 1, 2$. More quantitatively,

$$\liminf_{n \rightarrow \infty} n \max_{i=1,2} |\text{Var}(Z_{F_i,c}(n)) - 1| \geq \frac{|K_{F_1} - K_{F_2}|}{2}. \quad (161)$$

Proof. By definition $g_F^*(n) = C_F^2 / \text{Var}(L_n)$, so

$$\text{Var}(Z_{F,c}(n)) = \frac{n-c}{g_F^*(n)}. \quad (162)$$

Since $g_F^*(n) = n - K_F + O(n^{-1})$, series division gives

$$\frac{n-c}{g_F^*(n)} = 1 + \frac{K_F - c}{n} + O(n^{-2}). \quad (163)$$

For any real c , at least one of $|K_{F_1} - c|$ and $|K_{F_2} - c|$ is at least $|K_{F_1} - K_{F_2}|/2$, proving the bound. $\square$

10 Exact special-law corollaries

10.1 Laplace increments

For a centered Laplace law, $R = |\varepsilon|/\mathbb{E}|\varepsilon| \sim \text{Exp}(1)$, so

$$m_k = k!. \quad (164)$$

Substitution into Theorem 3 gives

$$\boxed{\text{Var}(L_n) = 4\alpha + \frac{16}{3}\alpha^2 + 6\alpha^3 + \frac{52}{9}\alpha^4 + O(\alpha^5).} \quad (165)$$

Since $C_L = 2$, the series-inversion corollary gives

$$\boxed{g_L^*(n) = n - \frac{4}{3} + \frac{5}{18n} + \frac{5}{27n^2} + O(n^{-3}).} \quad (166)$$

Equivalently, the variance-normalized coordinate may be written

$$Z_{L,n} = \frac{1}{2} \sqrt{n - \frac{4}{3} + \frac{5}{18n} + \frac{5}{27n^2} + O(n^{-3})} L_n. \quad (167)$$

10.2 Gaussian increments

For a centered Gaussian law,

$$m_2 = \frac{\pi}{2}, \quad m_3 = \pi, \quad m_4 = \frac{3\pi^2}{4}, \quad m_5 = 2\pi^2. \quad (168)$$

The variance coefficients become

$$v_1 = \pi, \quad (169)$$

$$v_2 = \frac{\pi(15\pi - 28)}{12}, \quad (170)$$

$$v_3 = \frac{\pi(16\pi^2 - 45\pi - 3)}{6}, \quad (171)$$

$$v_4 = \frac{\pi(2925\pi^3 - 11010\pi^2 + 5981\pi - 36)}{360}. \quad (172)$$

Since $C_G = \sqrt{\pi}$,

$$\boxed{g_G^*(n) = n - K_G + \frac{A_G}{n} + \frac{B_G}{n^2} + O(n^{-3})}, \quad (173)$$

where

$$K_G = \frac{15\pi - 28}{12} \approx 1.593657483654, \quad (174)$$

$$A_G = \frac{856 + 240\pi - 159\pi^2}{144} \approx 0.282744007558, \quad (175)$$

$$B_G = \frac{89220\pi^2 + 130784 - 28344\pi - 29475\pi^3}{8640} \approx 0.971620062501. \quad (176)$$

10.3 Rademacher increments: an independent coefficient check

Let $\varepsilon = \pm 1$ with probability $1/2$. Then $R \equiv 1$, hence

$$y \equiv 0, \quad m_2 = m_3 = m_4 = m_5 = 1, \quad (177)$$

and

$$L = 2 \operatorname{atanh}(x), \quad x = \sum_j w_j Q_j. \quad (178)$$

Theorem 3 predicts

$$\boxed{\operatorname{Var}(L_n) = 2\alpha + 3\alpha^2 + 5\alpha^3 + \frac{53}{6}\alpha^4 + O(\alpha^5)}. \quad (179)$$

This can be recovered independently without any x - y joint algebra. The Rademacher weighted sum has

$$\mathbb{E}x^2 = s_2, \quad (180)$$

$$\mathbb{E}x^4 = 3s_2^2 - 2s_4, \quad (181)$$

with higher even moments obtained from the Rademacher cumulants. Substitution into

$$\operatorname{atanh}^2 x = x^2 + \frac{2}{3}x^4 + \frac{23}{45}x^6 + \frac{44}{105}x^8 + O(x^{10}) \quad (182)$$

and the exact s_r expansions give the same four coefficients. Because $y \equiv 0$, this calculation avoids the joint x - y cumulant algebra and independently reproduces v_3 and v_4 .

11 Formal verification and reproducible checks

The accompanying Lean 4 source separates exact pathwise results, kernel algebra, weak-limit transfer, and distributional benchmarks. It is available at

<https://github.com/pawelmalmyga-ai/EndpointCoordinate>.

Table 3: Main mathematical layers represented in Lean.

Claim layer	Lean module	Representative declaration
Matched endpoint identity	Core	<code>matched_endpoint_identity,</code> <code>finite_path_geometry</code>
Classical Wilder up/down recursion bridge	Core	<code>directionalRun_matches_run,</code> <code>directionalRun_from_initial_matches</code>
RSI and log-odds chain	Core	<code>exact_rsi_logOdds_chain</code>
Recursive Wilder RSI/log-odds chain	Core	<code>wilder_directionalRun_rsi_logOdds_chain</code>
Wilder displacement bound	Core	<code>wilder_finite_path_displacement_interval</code>
Weight energy and effective length	Probability	<code>wilderWeight_sq_tsum,</code> <code>wilder_effectiveLength</code>
Stationary smoother and tail energy	Stationary	<code>stationaryWilderSeries_recursion,</code> <code>wilderWeight_sq_tail_fraction</code>
Local log-odds linearization	Asymptotics	<code>logOdds_isEquivalent_twice_id</code>
Weak-limit transfer	CLT	<code>effectiveLength_half_logOdds_clt_transfer</code>
Gaussian and Laplace benchmarks	Gaussian, Laplace, TriangularArray	<code>finiteGaussianWilderRow_clt,</code> <code>finiteLaplaceWilderRow_clt</code>
Finite-variance linear-row CLT	FiniteVarianceCLT	<code>finiteVarianceWilderRow_clt</code>
Student moment substitutions	Student	<code>student_firstVarianceShift_from_radial_moments</code>

The Lean sources contain no `sorry`, `admit`, or project-defined axioms. Kernel acceptance establishes each listed conclusion from the assumptions in its theorem type; it does not establish that those assumptions hold for market data.

The theorem `finiteVarianceWilderRow_clt` concerns normalized finite linear weighted rows with vanishing omitted squared-weight mass. The declarations in `CLT.lean` conditionally transfer an already established limit through the local log-odds map. Thus the stationary nonlinear theorem in Section 6 is proved conventionally in this paper rather than as one composed Lean theorem. The Berry–Esseen proof and the analytic localization and remainder estimates used for the fourth-order variance expansion are likewise outside the present formalization.

The repository’s pinned SymPy script exactly reproduces the finite coefficient enumeration and is run in continuous integration. This is an independent check of the algebra, not a substitute for the analytic estimates in Section 8.

12 Scope and limitations

The finite-horizon expansion uses stronger assumptions than the large- n Gaussian limit. The following distinctions delimit its interpretation.

1. **Symmetry is essential for the finite-horizon variance expansion.** It makes the sign independent of magnitude, forces odd- x cumulants to vanish, and gives $\mathbb{E}L = 0$. An asymmetric innovation law generally produces finite- n centering terms and different coefficient algebra. No asymmetric analogue is claimed here.
2. **The expansion theorem is light-tail.** The exponential-moment assumption is used for fixed-deviation concentration, while the logarithmic integrability condition controls rare events where one directional mass is very small. These assumptions cover Gaussian, Laplace, and bounded laws such as Rademacher. They do not cover Student- t laws.

3. **The expansion is not an exact distributional pivot.** $g_F^*(n)$ standardizes variance, while Theorem 2 controls convergence of the full distribution. The displayed formulas are asymptotic expansions of $C_F^2 / \text{Var}(L_n)$, not closed forms for the exact finite- n variance.
4. **The Student- t extension has a separate status.** The companion technical note proposes a first-order result for Student- t_ν , $\nu > 3$. Lean checks the moment substitutions and constants, but the heavy-tail analytic argument is not formalized and is not used in the theorems of this paper.
5. **Input scale matters for probabilistic interpretation.** The exact path identities hold for price and log-price inputs alike. The limit theorem and finite-horizon coefficients apply to i.i.d. increments of the chosen input. Under a multiplicative price path they apply directly to log-price RSI, not as exact formulas for classical price RSI; the latter comparison is reported only as a numerical robustness audit.
6. **No predictive statement is made.** The results concern the geometry and normalization of the RSI log-odds coordinate only.

13 Research provenance and computational assistance

The author formulated the research question, selected the mathematical framework and assumptions, and is responsible for the claims and their interpretation. AI tools assisted with proof drafting, Lean translation, symbolic checking, debugging, and editorial preparation. Lean verification applies only to the declarations identified in Section 11; the remaining arguments are conventional mathematics. The repository’s `PROVENANCE.md` provides a fuller disclosure.

14 Conclusion

Wilder’s directional-mass construction admits a natural unbounded coordinate

$$L_n = \log(U_n/D_n) = 2 \operatorname{atanh}(F_n). \quad (183)$$

The relation between this coordinate and price displacement has an exact finite-horizon geometry, while its stochastic scale has a separate probabilistic structure. Under zero-mean i.i.d. finite-variance increments,

$$\frac{\sqrt{2n-1} \mathbb{E}|\varepsilon|}{2\sigma} L_n \Rightarrow N(0, 1). \quad (184)$$

The equivalent $\sqrt{n}$ representation uses $C_F = \sqrt{2}\sigma/\mathbb{E}|\varepsilon|$. The factor $2n-1$ is the exact inverse-energy length of the linear Wilder kernel; it is not an exact finite- n normalization of the nonlinear log-odds coordinate. With a finite third absolute moment, the full nonlinear log ratio admits an $O(n^{-1/2})$ Berry–Esseen bound.

Under stronger symmetric light-tail assumptions, the variance itself has a controlled finite-horizon expansion through fourth order in $\alpha = 1/n$. That expansion yields a law-specific effective normalization length

$$g_F^*(n) = n - K_F + \frac{A_F}{n} + \frac{B_F}{n^2} + O(n^{-3}), \quad (185)$$

with analytically determined coefficients. In particular,

$$g_L^*(n) = n - \frac{4}{3} + \frac{5}{18n} + \frac{5}{27n^2} + O(n^{-3}) \quad (186)$$

for Laplace increments, while Gaussian increments produce a distinct π -dependent correction.

These law-specific formulas are exact asymptotic statements for the additive input model just described. When the chosen input is log price they apply directly to log-return innovations. Their behavior for classical price RSI after exponentiating log returns is a separate finite-sample question, treated numerically in the accompanying price-path audit rather than asserted as a consequence of the theorems above.

The exact geometry, its finite-path connection to the classical recursive Wilder masses, effective-length identities, conditional log-odds transfer, and the finite-variance linear-row limit theorem are checked in Lean. The stationary nonlinear limit, Berry–Esseen bound, and higher-order analytic estimates are conventional proofs, with their status summarized in Tables 1 and 3.

A Derivation of the one-observation cumulants

For $q = QR$ and $r = R - 1$, sign symmetry gives

$$\mathbb{E}[q^a r^b] = 0 \quad \text{for odd } a. \quad (187)$$

For even a ,

$$\mathbb{E}[q^a r^b] = \mathbb{E}[R^a (R - 1)^b] = \sum_{k=0}^b \binom{b}{k} (-1)^{b-k} m_{a+k}. \quad (188)$$

The cumulants in Table 2 follow from the standard partition formula

$$\text{cum}(Z_1, \dots, Z_r) = \sum_{\pi \in \mathcal{P}_r} (|\pi| - 1)! (-1)^{|\pi|-1} \prod_{B \in \pi} \mathbb{E} \prod_{i \in B} Z_i. \quad (189)$$

Only orders two through five are required because $s_r(\alpha) = O(\alpha^{r-1})$ and $r \geq 6$ therefore begins at $O(\alpha^5)$.

B Reproducible coefficient calculation

The coefficient calculation in Theorem 3 is finite and deterministic. It uses:

1. the polynomial P_8 displayed in the finite-horizon expansion section;
2. the exact cumulant relation $\text{cum}(x^{[a]}, y^{[b]}) = s_{a+b} c_{a,b}$;
3. the ordinary moment–cumulant partition formula;
4. the α expansions of s_2, s_3, s_4, s_5 .

An accompanying Python/SymPy script, `supplement/moment_expansion_audit.py`, enumerates set partitions exactly, uses symbolic rational arithmetic, and returns

$$\begin{aligned} v_1 &= 2m_2, \\ v_2 &= (15m_2^2 + 2m_2 - 8m_3)/3, \\ v_3 &= (64m_2^3 + 5m_2^2 - 64m_2m_3 + m_2 - 2m_3 + 11m_4)/3, \end{aligned}$$

and the displayed v_4 . It also reproduces the Laplace, Rademacher, Gaussian, and Student-law substitutions used in the repository. The script performs no simulation and contains no fitted coefficient. It is run in continuous integration. Its role is algebraic; the analytic remainder estimates are supplied by Section 8.

C Relation to the earlier empirical scaling

The earlier practitioner study [4] used the compact rule $2/\sqrt{n-1}$ as an empirically successful cross-lookback scale and explicitly identified formal mathematical validation as an open problem. The present analysis separates three objects that should not be conflated:

1. the exact geometric support scale $n - 1$ of the endpoint displacement;
2. the exact inverse-energy length $2n - 1$ of the linear Wilder kernel;
3. the law-dependent finite- n variance normalizer $g_F^*(n)$ of the full nonlinear log-odds coordinate.

The practical $2/\sqrt{n-1}$ rule is therefore neither the exact kernel-energy normalization nor a universal finite- n theorem. For Laplace increments, $C_F = 2$ and $g_L^*(n) = n - 4/3 + O(1/n)$, so the earlier expression is close but not exact. For Gaussian increments, both the law constant and the finite correction differ. Its practical performance remains an empirical question, separate from the exact and asymptotic results proved here.

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